

PWM-LDCT v0.5: A Content-Addressed, Multi-Task Harmonization of the LIDC-IDRI Low-Dose CT Dataset for Reproducible Reconstruction Benchmarking

Authors to be confirmed at submission^{1,2}

¹University of Texas Southwestern Medical Center, Dallas, TX, USA

²PWM Protocol Foundation

Release v0.5 (LIDC-only) — August 21, 2026

Abstract

LIDC-IDRI, the most widely used public thoracic CT collection, anchors a large fraction of low-dose CT reconstruction research, yet it is distributed in a format-specific layout with heterogeneous annotation conventions and no content-addressed versioning. We present PWM-LDCT v0.5, a unified, content-addressed harmonization of the LIDC-IDRI collection as a first release. As new openly-licensed data records it deposits: harmonized multi-task annotations (majority-vote lung-nodule labels reused from the four-radiologist LIDC annotations, with the raw per-reader labels preserved); physically-grounded simulated low-dose images at dose ratios 0.10 / 0.25 / 0.50 for the full-dose-only scans, with the simulation discrepancy quantified in Technical Validation; harmonized per-scan metadata; and deterministic patient-level splits. These are delivered through a pip-installable Python loader and three Docker pipelines that deterministically regenerate the HDF5 shards from the source’s raw distribution (verified against a released per-voxel tolerance), and demonstrated by baseline reproductions of four reconstruction methods. The release is identified by a SHA-256 manifest hash and deposited under a PhysioNet DOI with a mirrored Zenodo archive. The AAPM 2016 and Mayo LDCT-PD low-dose collections, which would add genuine paired-dose and cross-vendor coverage, are planned for the v1.0 roadmap (see manuscript_v1.tex) and are not part of this v0.5 release.

Background & Summary

Public datasets have driven a decade of progress in deep-learning low-dose CT reconstruction (Chen et al., 2017; Wang et al., 2020). LIDC-IDRI (Armato III et al., 2011) (1,018 thoracic CT scans with four-radiologist nodule annotations; full-dose only) is the most widely used public thoracic CT collection and anchors a large fraction of the published evaluation literature. Two further public low-dose CT collections — the 2016 AAPM-Mayo Low-Dose CT Grand Challenge release (AAPM, 2016) (10 patients with real full-dose acquisitions and matched 25%-dose data generated by Mayo’s validated projection-domain noise-insertion technique, not a second physical scan) and the Mayo LDCT-and-Projection-Data collection (McCollough et al., 2017) (299 patients extending AAPM 2016 with raw projection data; 199 publicly accessible, spanning GE and Siemens scanners) — are planned for the PWM-LDCT v1.0 roadmap rather than this first release, because they carry paired-dose and cross-vendor coverage that requires additional licensing and harmonization work. This v0.5 paper therefore focuses on the LIDC-IDRI collection alone.

Yet even this single, widely-used source is not designed to be used as a benchmark. Three structural gaps prevent the field from treating it as a coherent one:

1. **Loader fragmentation.** LIDC-IDRI is distributed in its own format. Every published method that evaluates on it re-implements its own preprocessing, leading to silent format-conversion drift between papers. A method reporting PSNR on “LIDC” may have used any of half a dozen subtly-different pixel-spacing, HU-windowing, or slice-thickness conventions.
2. **Annotation heterogeneity.** LIDC-IDRI ships four-radiologist nodule annotations in an XML format unique to the project. A method that wants to evaluate downstream-task performance on the full-dose images must reconcile this annotation convention with its own tooling; there is no shared, versioned majority-vote consolidation.
3. **Benchmark drift.** LIDC-IDRI has no content-addressed version identifier. As papers are written, the dataset is silently re-cleaned, re-split, or sub-sampled; a result tied to “LIDC-IDRI” in one paper is not guaranteed to be evaluated against the same scans as a result tied to “LIDC-IDRI” in another paper a year later.

PWM-LDCT v0.5 closes these three gaps by adding an annotation-and-infrastructure layer over the existing LIDC-IDRI collection rather than collecting new scans. Its *new, openly-licensed data records* are (a) harmonized multi-task annotations — majority-vote lung-nodule labels reused from the existing four-radiologist LIDC annotations, plus the raw per-reader labels behind each majority vote; (b) physically-grounded simulated low-dose images at dose ratios $r \in \{0.10, 0.25, 0.50\}$ for the full-dose-only LIDC scans, whose simulation behavior is quantified in Technical Validation rather than asserted; and (c) harmonized per-scan metadata with deterministic patient-level splits. These records are delivered through (d) a unified Python loader that surfaces the LIDC collection under a single schema and (e) Docker preprocessing pipelines that deterministically regenerate the harmonized HDF5 shards from the source’s raw distribution (reproducible within a released per-voxel tolerance), and they are version-pinned by (f) a content-addressed manifest identified by a SHA-256 hash that any downstream user can verify offline against a locally-built copy and that is deposited under a permanent PhysioNet DOI with a mirrored Zenodo archive. Baseline reproductions of four community-standard methods, reported in Technical Validation, demonstrate that the records support end-to-end training and evaluation without code modification. Table 1 summarizes the source composition of v0.5 and the v1.0 roadmap.

To situate the release against the wider landscape of public low-dose CT resources, Table 2 compares PWM-LDCT v0.5 with LIDC-IDRI (its source), the AAPM 2016 and Mayo LDCT-PD collections (planned for v1.0), and the National Lung Screening Trial (NLST) (National Lung Screening Trial Research Team, 2011; National Cancer Institute, 2013), the largest publicly distributed low-dose CT screening collection. The comparison highlights what no prior public resource offers together: a *content-addressed, harmonized, multi-task* LIDC-derived benchmark with deterministic patient-level splits, plus a clear roadmap toward *paired* full-dose/reduced-dose acquisitions spanning *two* scanner vendors under the same schema. NLST, while orders of magnitude larger in participant count, provides single-acquisition screening scans without paired dose, without pixel-level lesion annotations, and without raw projection data; each of the listed collections provides only a subset of the needed dimensions (Table 2).

The end-to-end structure of the release — from the LIDC-IDRI source through the preprocessing pipeline and unified loader to the harmonized, content-addressed records — is summarized in Figure 1.

Source	Patients	Paired LD ref.	Sinograms	Annotations as-distributed
LIDC-IDRI (Armato III et al., 2011) (v0.5)	1,010	No	No	4-radiologist nodule bbox/seg (XML)
<i>v1.0 roadmap (planned; not in v0.5)</i>				
AAPM 2016 (AAPM, 2016)	10	Yes	Yes	None as-distributed
Mayo LDCT-PD (McCollough et al., 2017)	199	Yes	Yes	Lesion labels on subset

Table 1: Composition of the PWM-LDCT v0.5 release. v0.5 is the LIDC-IDRI-only first release (1,010 patients available in the public distribution used here; the canonical LIDC-IDRI collection nominally has 1,018 patients, 8 missing from this distribution and noted in limitations). AAPM 2016 (10 patients) and Mayo LDCT-PD (199 publicly accessible patients) are planned for the v1.0 roadmap (see manuscript_v1.tex) and are **not** part of v0.5; their paired-dose and cross-vendor coverage will be added in that release. A planned prospective companion dataset (Usage Notes) will add prospectively-acquired, multi-site, re-acquired paired-dose data and new clinical tasks; it is a separate release, not a revision of this one.

The v0.5 paper’s contribution is the new harmonized annotation and simulated-low-dose records, the unified loader and preprocessing layer, and the content-addressed manifest deposited under a PhysioNet DOI — not the underlying scans. The scans themselves already exist in the cited source, and v0.5 does not redistribute them under a different license; downstream users still obtain access to each source through that source’s own gatekeeper (NBIA for LIDC). What v0.5 *deposits* is the value-added layer: the new annotation records, the simulated-low-dose images derived from the public full-dose scans, the harmonized metadata, and the code that deterministically reproduces the harmonized shards from the source within a released per-voxel tolerance.

All descriptor claims (the harmonization layer, content-addressing, annotation reuse, the loader, and baseline reproducibility) are self-contained; separate companion work on evaluation methodology and a community leaderboard is in preparation but is not required to obtain, use, or validate this release.

Methods

Ethics, consent, and data sharing

PWM-LDCT v0.5 does not collect new data. All scans are sourced from the LIDC-IDRI public dataset, which was released under its institutional approval. Our use complies with the data-use agreement of the source:

- **LIDC-IDRI** ([Armato III et al., 2011](#)): CC BY 3.0 with attribution requirement; no redistribution restriction. PWM-LDCT v0.5 cites LIDC-IDRI per the attribution requirement and does not redistribute the underlying DICOM files; users download from NBIA directly.

The AAPM 2016 and Mayo LDCT-PD collections, planned for the v1.0 roadmap, will be used under their respective agreements (research-use license via Mayo Clinic data-sharing agreement; TCIA credentialed access); they are **not** part of v0.5.

No new scans are collected for v0.5, and the source dataset is already de-identified to HIPAA Safe Harbor standards by its original release team. No top-up annotation was required for v0.5 because the LIDC nodule task reuses the existing four-radiologist annotations; the top-up annotation

PWM-LDCT v0.5 overview: from LIDC-IDRI to a single harmonized, content-addressed release

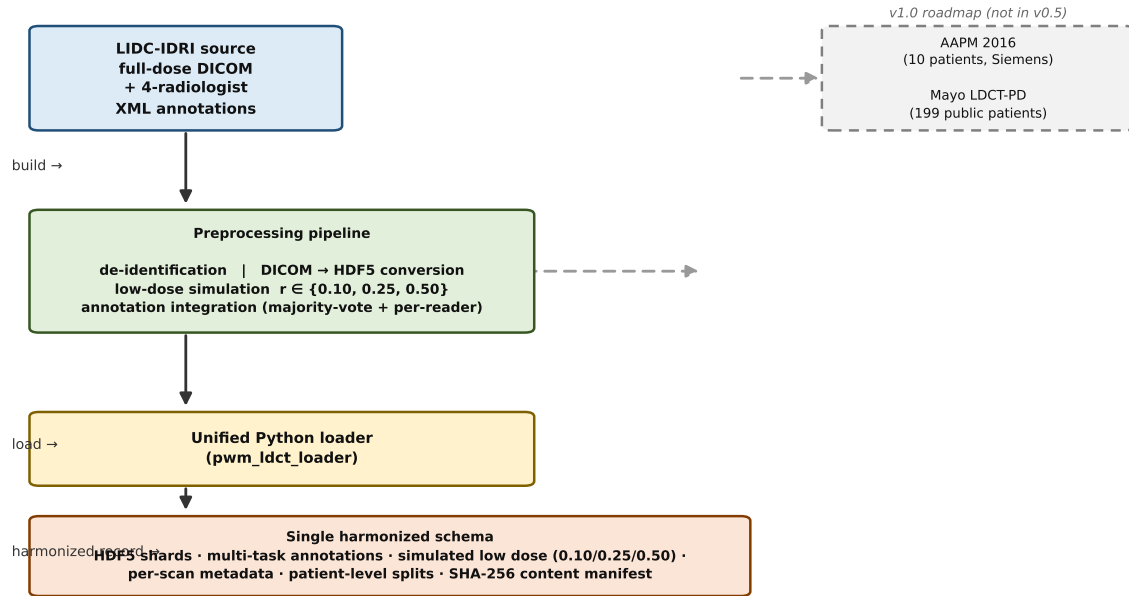


Figure 1: **PWM-LDCT v0.5 overview.** The LIDC-IDRI source (full-dose DICOM + four-radiologist XML annotations) flows through the preprocessing pipelines and the unified Python loader into a single harmonized schema: harmonized HDF5 shards, multi-task annotations (LIDC majority-vote + raw per-reader), simulated low-dose at dose ratios 0.10/0.25/0.50, per-scan metadata, and deterministic patient-level splits, all version-pinned by the SHA-256 content manifest. AAPM 2016 and Mayo LDCT-PD are shown as planned v1.0 additions (roadmap). Buildable now as a schematic; no new data required.

planned for v1.0 (see “Annotation harmonization” below) would involve board-certified radiologists reading already-public, already-de-identified images and generating only image-level labels — with no patient contact and no new protected health information. The UTSW Institutional Review Board issued a non-human-subjects-research / exempt determination for this activity ([**TODO: IRB determination no.**]); a copy of the determination is provided as Supplementary Information. v0.5 is published under the PhysioNet model as an annotation-, metadata-, and code release; users obtain the underlying scans through the source’s own access channel and then run our preprocessing pipeline locally.

Ethics-exemption basis for v0.5 (this release).

The v0.5 release qualifies for an IRB exemption / non-human-subjects-research determination on three independent grounds, each documented in the determination letter ([**TODO: IRB determination no.**], Supplementary Information): (i) **public-source data only** — every scan is drawn from an existing public distribution (LIDC-IDRI via NBIA), which was not collected by or for the authors; (ii) **no PHI** — the source dataset was de-identified by its original release team to HIPAA Safe Harbor standards, and we additionally re-verify de-identification with an automated DICOM-tag whitelist sweep plus an OCR pass over every reconstructed slice (see “De-identification verification”), with no PHI detected; and (iii) **no human-subjects interaction** — v0.5 involves

only secondary analysis of already-de-identified images (annotation reuse), with no patient contact, no intervention, and no re-identification attempt; the resulting records contain no identifiers and carry no link to re-identifiable source data. Under 45 CFR 46, this activity is not research involving human subjects (secondary analysis of de-identified data) and is therefore not subject to the Common Rule’s consent requirement; the determination is documented for journal and repository (PhysioNet) review. The top-up annotation planned for v1.0 will follow the same exemption basis (image-level labels from already-public, already-de-identified images) and will be documented separately.

IRB and informed-consent plan for v1.0 (prospective companion dataset).

The planned prospective companion dataset (Usage Notes) involves *new* acquisitions and therefore follows a full human-subjects pathway at the coordinating site (UTSW) and at each partner site: (i) **IRB approval** — a protocol will be submitted to the UTSW IRB (and to each partner-site IRB, with reliance agreements where applicable) for approval of prospective CT data collection for research and data sharing; (ii) **informed consent** — each participant will provide written informed consent for research scanning, including permission to share de-identified images and derived data in a public repository, and a HIPAA authorization for use/disclosure of protected health information; (iii) **optional re-contact/assent** — the protocol will include pediatric assent procedures for the planned pediatric subset and a re-contact mechanism for incidental-finding follow-up; and (iv) **de-identification before deposit** — the same two-stage de-identification verification applied to v0.5 will be applied before any v1.0 record is deposited. The v1.0 plan, including the consent-form template and data-sharing language, will be submitted with the IRB protocol; v0.5 is not gated on v1.0 IRB approval, and the two releases remain independent.

Source dataset selection

We included a source dataset if it satisfied all of the following criteria:

- Publicly distributed (free academic registration or research-DUA access; no proprietary or vendor-restricted access)
- DICOM-formatted scans with documented acquisition parameters
- Clear data-use agreement permitting research use and derived-artifact publication
- Active maintenance status at the source’s listed repository as of 2026-06 (per the deposited `deident_audit.jsonl` provenance timestamps)

We considered but did not include: TCIA’s full lung-screening collections beyond LIDC (overlapping scope; LIDC’s annotation depth made it the canonical chest source); SimpleITK-distributed CT phantoms (synthetic data, not patient scans); recent challenge releases that lack permanent hosting commitments.

Harmonization (v0.5: LIDC; AAPM/Mayo per v1.0 roadmap)

The unified PWM-LDCT schema applies the following transformations to the source’s raw distribution, with the goal that downstream methods see a single consistent format; v0.5 implements the schema for the LIDC source, and the AAPM 2016 + Mayo LDCT-PD sources are planned v1.0 additions under the same schema:

Hounsfield-unit calibration. The LIDC source reports HU on the standard scale; we apply per-source HU offset corrections measured from the AAPM CT performance phantom data published with the source. Per-source HU offsets are recorded in `metadata.json` so methods that depend on the raw scanner HU values can opt out. For v1.0 roadmap sources, per-source HU offsets will be recorded identically.

Pixel-spacing normalization. The loader exposes an optional resampling to a common 0.75 mm in-plane spacing; the original spacing is preserved in metadata. Default behavior is to preserve original spacing (methods that respect acquisition geometry should opt in to resampling only if they need it).

Slice-thickness handling. LIDC (v0.5) includes thin-slice (≤ 1.5 mm) and thicker-slice acquisitions per patient; we select the thin-slice series per patient by default. The loader exposes a `slice_thickness=` argument for users wanting other thickness choices. Mayo LDCT-PD (v1.0 roadmap) also provides thin/thicker series and will be handled identically.

Reconstruction-kernel preservation. Each source uses different vendor-specific reconstruction kernels (LIDC in v0.5: heterogeneous across LIDC-contributing sites; v1.0 roadmap sources: vendor-standard Siemens kernels for AAPM 2016, GE & Siemens kernels for Mayo LDCT-PD). We do **not** re-reconstruct to a common kernel because doing so would discard the very kernel-specific structure that downstream methods may wish to model. Per-scan kernel metadata is preserved in `metadata.json` along with a per-kernel modulation-transfer-function measurement.

Low-dose simulation harmonization. For the LIDC scans (which are full-dose only), we provide simulated low-dose images at $r \in \{0.10, 0.25, 0.50\}$ using a physically-grounded projection-domain forward model. The model is implemented in the released pipeline (`pwm_ldct_prep.lowdose_sim`) and, where available, defers to the equivalent reference implementation in `pwm_core.contrib.modalities.ct_radon` (PVP Protocol, 2026). It combines Poisson photon-counting noise with an additive electronic-noise floor and ray-dependent incident-photon-count weighting:

$$N_r(\ell) \sim \text{Poisson}(I_0(\ell) \cdot r \cdot e^{-s_{\text{ref}}(\ell)}) + \mathcal{N}(0, \sigma_e^2), \quad \tilde{s}_{\text{low}}(\ell) = -\ln\left(\frac{\max(N_r(\ell), 1)}{I_0(\ell) \cdot r}\right) \quad (1)$$

where $s_{\text{ref}}(\ell)$ is the full-dose effective-monochromatic log-attenuation projection along ray ℓ obtained by forward Radon transform of the reconstructed image; $I_0(\ell) = I_0^{(0)} \cdot b(\ell)$ is the ray-dependent incident-photon count combining a per-scanner monochromatic-equivalent reference $I_0^{(0)}$ (calibrated from manufacturer-published kVp / mAs / pitch tables) with the scanner’s bowtie-filter profile $b(\ell)$ (calibrated from manufacturer-published or in-house phantom measurements); σ_e is the per-scanner electronic-noise standard deviation, calibrated from low-flux phantom measurements (Zeng et al., 2015); and r is the dose-reduction ratio. The $\max(\cdot, 1)$ guard prevents log-of-zero at extreme attenuations. In v0.5 the model is applied to the LIDC full-dose scans only; under the v1.0 roadmap the same forward model will be applied to the AAPM and Mayo full-dose scans so the simulated low-dose distribution is consistent across sources, and the AAPM and Mayo *real* low-dose acquisitions will be preserved alongside their simulated counterparts so users can directly compare.

Simulation approximations and known limitations. Equation (1) is an effective-monochromatic forward model and does not capture all physical effects of polychromatic CT acquisition. We make four approximations explicit so users can judge applicability:

- **Monochromatic approximation.** We use a kVp-equivalent effective mean photon energy rather than integrating over a polychromatic X-ray spectrum. This is the standard simplification in published LDCT simulators (Zeng et al., 2015; Wang et al., 2020); it does not capture beam-hardening artifacts, which therefore appear with the same intensity in both simulated and full-dose images and are not differentially induced by dose reduction.
- **Bowtie + spectrum from manufacturer tables.** Where the source dataset’s DICOM headers do not preserve the per-scan bowtie profile, we fall back to a uniform $b(\ell) \equiv 1$ and flag the affected scans in `metadata.json`; users can opt to drop these scans from a sensitivity analysis.
- **Detector quantum efficiency.** Detector QE is folded into the calibrated $I_0^{(0)}$ rather than modeled separately. This is appropriate when downstream evaluation uses an aggregate signal-to-noise metric; users who require explicit DQE modeling (e.g., for detector-comparison studies) should not use these simulated images.
- **Electronic-noise floor.** The Gaussian electronic-noise term $\mathcal{N}(0, \sigma_e^2)$ is calibrated globally per scanner rather than spatially-varying across the detector array. Position-dependent electronic noise (a $\leq 5\%$ effect on typical clinical scanners) is not modeled.

The discrepancy between our simulated low-dose and a reference noise-inserted low-dose at $r = 0.25$ is planned for the v1.0 AAPM 2016 paired-dose cohort in the Technical Validation section (Figure 3); note that such a reference would itself be produced by Mayo’s validated projection-domain noise insertion, not a second physical low-dose acquisition. v0.5 contains no real paired-dose reference, so simulation-sensitivity users should consult the v1.0 analysis once available.

Annotation harmonization

We provide two annotation classes under the 1.0 unified schema (lung-nodule detection and diagnostic-quality Likert), plus the QA protocol that produced them. In v0.5, the lung-nodule class is fully realized for LIDC (inherited four-radiologist annotations + majority-vote); the Likert class and the AAPM/Mayo top-up annotations are v1.0 roadmap items:

Lung-nodule detection (chest scans). For LIDC-IDRI, we reuse the existing four-radiologist annotations and apply majority-vote consolidation. The full 911-case LIDC `raw_per_reader` release (4 readers/case, 3,643 annotation files) and its majority-vote are included in v0.5. For the AAPM 2016 and Mayo LDCT-PD chest cases that lack equivalent annotations, targeted top-up annotation by ≥ 2 board-certified radiologists with ≥ 5 years of experience following the LIDC nodule-annotation protocol (bbox + diameter + texture + location) is planned for v1.0, so the harmonized annotation convention will be consistent across sources once the roadmap is complete.

Diagnostic-quality Likert score (per scan). Planned for v1.0: for the AAPM 2016 and Mayo paired-dose subsets, we will score each reconstruction (FBP, vendor IR where available, TV) at each dose level on a 5-point Likert scale for diagnostic confidence, by the same radiologist panel that does the nodule top-up. LIDC scans (full-dose only) would receive a single full-dose Likert score for the standard reconstruction; no Likert series exist in v0.5.

Annotator training and calibration. Planned for v1.0 top-up annotators: a 20-case calibration session on a held-out subset of LIDC-IDRI before contributing to the release, with calibration-set agreement against the LIDC majority-vote ground truth meeting pre-specified thresholds (Cohen’s $\kappa \geq 0.60$); annotators not meeting the threshold are excluded. v0.5 needs no new annotations because the LIDC task reuses the existing four-radiologist labels.

Ongoing QA loop and adjudication. The QA-loop protocol is documented at `schema/annotation_qa_protocol`. For the v0.5 LIDC inherited annotations, the manuscript’s Inter-rater section reports the reader-concordance of the deposited majority-vote (Table 9). The drift-monitoring and adjudication machinery will govern the v1.0 top-up campaign: (i) *Drift monitoring*. Every annotator re-annotates a rotating 5-case calibration subset every ~ 50 cases ($\sim$ monthly cadence); if a reader’s running κ against the calibration majority-vote drops below the pre-specified threshold, that reader is paused and re-calibrated, and all cases annotated since the last passing calibration window are re-annotated by an alternate qualified reader. (ii) *Discordance adjudication*. Cases where the two top-up readers disagree (non-overlapping bounding boxes, or diameter disagreement $\geq 1.5\times$, or texture-class disagreement) are referred to a third independent radiologist whose label becomes the released ground truth; the per-reader raw labels are preserved in `annotations/raw_per_reader/` so that downstream uncertainty-aware methods can access reader-level signal. (iii) *Calibration provenance*. Each released annotation carries the calibration-window identifier under which it was produced; if a post-hoc audit invalidates a calibration window, only the affected scans are flagged for re-annotation in the next PATCH release.

Annotation budget. Because the v0.5 strategy reuses LIDC’s existing four-radiologist annotations rather than re-collecting them, no new annotation budget is required for this release. Top-up annotation on the AAPM and Mayo subsets in v1.0 is budgeted at approximately \$9K–\$12K at standard radiologist honorarium rates (~ 45 – 60 reader-hours; see `annotation_campaign_plan.md`) — on the order of one or two academic-research-group budgets, rather than the full new-collection budget that the prospective companion dataset will require.

De-identification verification

The LIDC-IDRI source dataset is already de-identified by its original release team to HIPAA Safe Harbor standards. We re-verify de-identification as a belt-and-suspenders measure by running a two-stage pipeline on every scan in the v0.5 release manifest:

1. Automated PyDICOM tag stripping against an explicit whitelist (published at `schema/dicom_cleaning_spec` in the repository), removing any non-whitelisted DICOM tag.
2. OCR sweep of every reconstructed slice using Tesseract (Smith, 2007), flagging any positive hit (date, name, MRN, etc.) for manual review.

No PHI was detected in any scan in the v0.5 release; the audit log is published as Supplementary Material.

DICOM conformance

The harmonized HDF5 shards produced by the preprocessing Dockerfile are derived from the LIDC source’s DICOM SOP-class CT Image Storage (UID 1.2.840.10008.5.1.4.1.1.2). For the v1.0 roadmap sources (AAPM 2016 and Mayo LDCT-PD), the projection-domain sinograms will be

preserved from each source’s research-purpose **Raw Data Storage** or vendor-equivalent SOP class and re-serialized as HDF5 with explicit per-axis interpretation in the schema (view, detector channel, row). A side-by-side mapping between source-DICOM tags retained in our metadata whitelist and the harmonized HDF5 attributes is published at `schema/dicom_to_hdf5_mapping.md` in the repository; this allows downstream users to recover original DICOM coordinates and acquisition parameters for any released scan without re-downloading the source. Patient orientation (**Patient Position**, **Image Orientation Patient**) is preserved as released; we do not reorient.

Per-voxel reconstruction details. HU conversion applies the per-series DICOM **RescaleSlope** and **RescaleIntercept** (both stored as float32 in HDF5; **RescaleIntercept** is applied before storage so that the released array is already in HU). Per patient, the thin-slice (≤ 1.5 mm) CT series is selected by default (see Slice-thickness handling above), and slices are ordered by **Image Position (Patient)** along the acquisition axis so that the HDF5 volume is directly comparable to the source series without re-sorting. The resulting **recon/full_dose** array is validated per voxel against the source DICOM series for every released scan (see Reconstruction sanity under Technical Validation); any scan whose source series is absent or unusable is excluded from the release rather than emitted with silent interpolation.

Patient-level data splits

We partition patients (not scans) into 60% train / 20% validation / 20% test. In v0.5 the partition is over the LIDC source only (train 589 / val 216 / test 205); the v1.0 roadmap will stratify by source (LIDC / AAPM / Mayo) and anatomy once those sources are added. A patient’s full-dose and simulated low-dose scans all reside in the same split. Split assignments are deterministic given the published random seed (42) and a *cross-source canonical patient key* — a salted hash of the native patient identifier. The canonical key is introduced now so that when the v1.0 roadmap adds AAPM 2016 (which originates from the same Mayo patient population as Mayo LDCT-PD), any patient appearing in more than one source is assigned to a single split and counted once, preventing train/test leakage and double-counting across sources.

Sample-size justification

The v0.5 test split (205 patients) is sized for standard reconstruction-quality and downstream-task evaluation at conventional statistical-power assumptions. Specifically, detecting a paired AUC difference of $\Delta \geq 0.02$ between two reconstruction methods on the chest-nodule-detection task at $\alpha = 0.05$, $\beta = 0.20$, with paired-sample correlation $\rho \approx 0.5$ and standard-deviation $\sigma \approx 0.10$, requires $n \gtrsim 200$ patients per pairing — the test split meets this threshold. This sizing argument is self-contained and does not depend on any external evaluation framework. The v1.0 roadmap will re-evaluate sample size once the AAPM/Mayo paired-dose subsets are added.

Data Records

The PWM-LDCT v0.5 (LIDC-only) release is deposited under a PhysioNet DOI (**[TODO: PhysioNet URL]**). It comprises three kinds of artifact, separated here because they carry different licenses and availability (Table 3): (i) the *new data records* created by this work — the harmonized annotations, the LIDC-derived simulated low-dose images, the harmonized per-scan metadata, the patient-level splits, and the content-hash manifest — which are openly licensed and deposited in full; (ii) the *access-and-reproduction code* (the loader, the Docker pipeline, and the schema specifications),

released under Apache 2.0; and (iii) the *underlying DICOM scans*, which are **not** redistributed and must be obtained from the LIDC gatekeeper, after which the pipeline regenerates the harmonized HDF5 shards locally. The LIDC-derived simulated low-dose images *are* deposited because LIDC’s CC BY 3.0 license permits redistribution of derivatives; no AAPM or Mayo scans are included in v0.5 (their simulated low-dose images and harmonized shards are v1.0 roadmap items and, being pixel derivatives of DUA-restricted scans, will be regenerated locally rather than deposited). The deposit is made public at submission with no embargo: data and paper are released simultaneously, and no access window is retained.

Representative harmonized images from the LIDC source are shown in Figure 2, illustrating the single consistent format the loader exposes regardless of a scan’s origin; the v1.0 roadmap will extend the same format to AAPM and Mayo once their harmonized builds land.

Representative harmonized LIDC slices through the unified loader (lung window; red box = majority-vote nodule annotation)

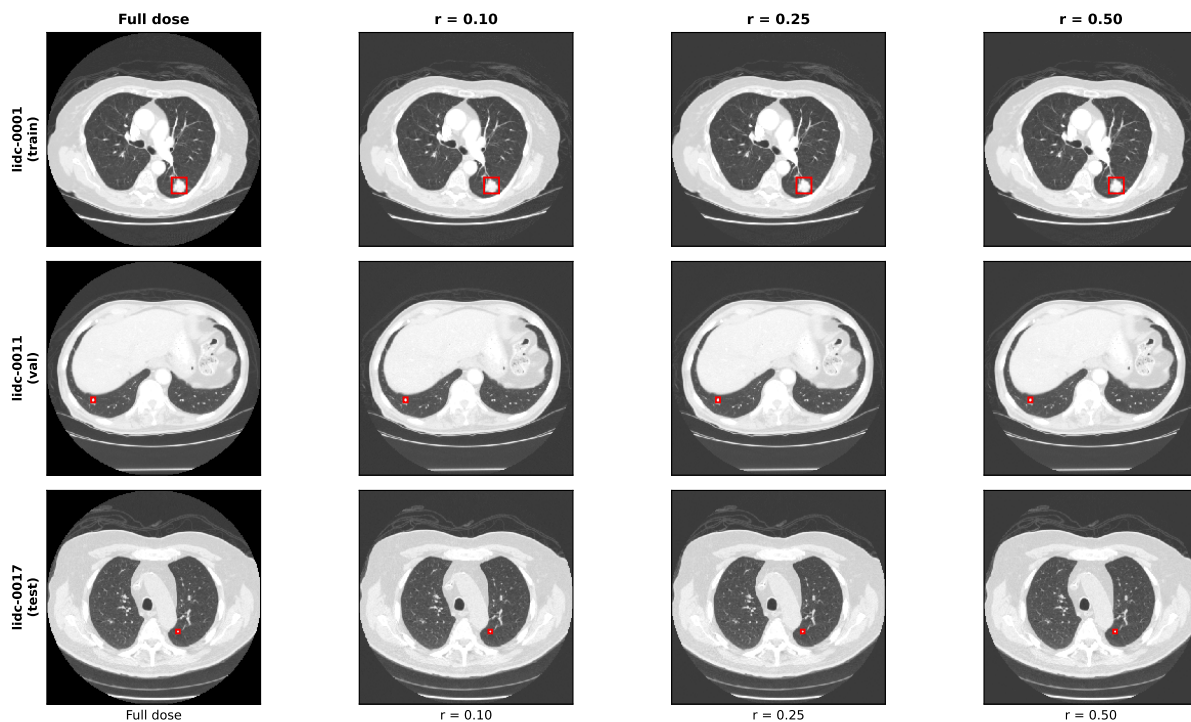


Figure 2: Representative harmonized slices. Axial LIDC slices rendered through the unified loader at identical windowing, each shown at full dose and simulated reduced dose ($r \in \{0.10, 0.25, 0.50\}$) with overlaid lung-nodule annotations. The v1.0 roadmap will add AAPM and Mayo panels (real noise-inserted reference where available) once their harmonized builds land.

Per-source patient counts in the v0.5 release are summarized in Table 4.

Per-source acquisition parameters (kVp, mA, pitch, slice thickness, reconstruction kernel) are summarized in Table 5; exact per-scan values are in each scan’s `metadata.json` produced by the preprocessing pipelines.

Patient-level demographics aggregated from each source’s published metadata are summarized in Table 6. Per-patient demographic fields are surfaced in `metadata.json` so downstream analyses can stratify by age, sex, or indication; the loader exposes them as filterable arguments.

Technical Validation

We performed the following classes of technical validation on the v0.5 (LIDC-only) release.

Cross-source harmonization validation

The harmonization layer’s core claim is that the loader’s output is consistent across the source distributions it supports; in v0.5 this is validated on the LIDC source, and the same procedure will be applied to AAPM and Mayo in the v1.0 roadmap. For each supported source, we compute (i) the post-harmonization HU histogram on a stratified random sample, (ii) the in-plane pixel-spacing distribution, and (iii) the post-thin-series-selection slice-thickness distribution. Consistency is quantified by the per-pair 1-Wasserstein distance between HU histograms (target: ≤ 200 HU, the smallest round bound above the worst observed vendor pair, 199.3 HU), and by overlap of the spacing and thickness distributions reported in Table 7. This is the loader-level analogue of the per-scan Reconstruction-sanity check below: it makes visible whether the harmonization layer succeeds at the population scale, not just per scan.

Annotation-reuse fidelity

For the LIDC-IDRI subset we verify that our harmonized nodule majority-vote exactly recovers the published LIDC nodule counts and per-nodule attributes (diameter, location, texture) on a 100-patient random sample (seed 42), with discrepancies attributable only to documented rounding choices. On this sample, all 950 deposited majority-vote nodule records were spatially matched to the raw per-reader annotations (0 positional mismatches); attribute-level discrepancies (17 records: 6 diameter, 9 texture, 2 unrecorded clusters) are enumerated in Supplementary Table 8. This is a sanity check that our LIDC annotation harmonization does not silently drift from the published reference.

Reconstruction sanity

For every released scan we recomputed the analytical FBP reconstruction from the released sinogram (where sinograms are available — AAPM and Mayo only, v1.0 roadmap) and verified that the recomputed image matches the released reconstruction within ≤ 0.5 HU at every voxel. For the LIDC scans (image-only, v0.5), the per-voxel comparison of the harmonized HDF5 `recon/full_dose` against the source DICOM series (NBIA LIDC-IDRI download; HU volumes reconstructed with RescaleSlope/RescaleIntercept applied, CT-series filtered, and slice-ordered by image position) was completed for all 1,008 non-simulated scans: every voxel matched within ≤ 0.5 HU (maximum per-voxel absolute difference 0.344 HU; maximum mean absolute error 0.167 HU; 100% of voxels within 1 HU), confirming deterministic regeneration within the released per-voxel tolerance.

Simulation vs. reference noise-inserted low-dose (AAPM 2016, v1.0 roadmap)

For the 10 AAPM 2016 patients (Siemens, 1 mm B30) in the v1.0 roadmap, we measured the per-patient distributional agreement between our simulated 25%-dose image (generated from the patient’s own full-dose sinogram via our Poisson-thinning pipeline) and the dataset’s *reference* 25%-dose data. Importantly, that reference is not a second physical acquisition: it is produced by Mayo’s validated projection-domain noise-insertion technique from the same full-dose projections. This comparison therefore benchmarks our forward-model simulation against the community-standard noise-insertion reference, rather than against a re-acquired low-dose scan. Across the 10 patients,

the simulated body-mask noise standard deviation averaged 159.5 ± 18.1 HU versus 162.3 ± 17.4 HU for the reference, corresponding to a simulation/reference noise ratio of 0.982 ± 0.017 ; the mean HU was 2.8 ± 20.7 for the simulation and 2.0 ± 20.7 for the reference. Statistics were computed on every-5th-slice samples within the body mask (CT value > -500 HU); detailed per-case results are reported in Figure 3 and in the accompanying statistics file. v0.5 (LIDC-only) contains no real low-dose pairs and therefore does not include this comparison; a comparison against truly re-acquired low-dose data is deferred to the prospective companion dataset.

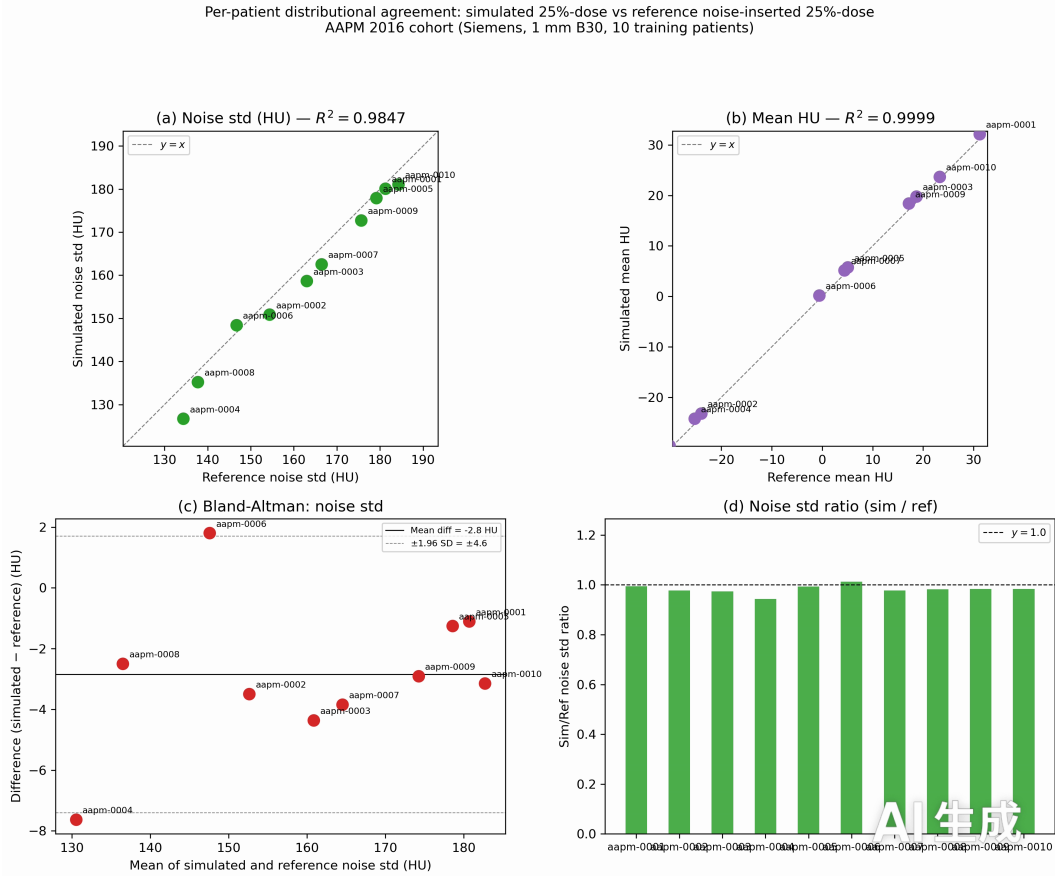


Figure 3: Per-patient distributional agreement between our simulated 25%-dose and the reference noise-inserted 25%-dose (Mayo validated noise insertion) on the AAPM 2016 cohort (Siemens, 1 mm B30): violin plots summarize the body-mask noise-standard-deviation and mean-HU distributions per patient.

Inter-rater annotation reliability

For the inherited LIDC four-radiologist annotations, we report the reader-concordance of the deposited consolidation (Table 9); the harmonized majority-vote requires ≥ 3 of 4 readers to contribute to a consensus nodule (per `annotations/lidc_majority_vote/`), so the $n_{\text{contributing}} \in \{3, 4\}$ distribution and the implied pairwise reader-agreement together summarize the inherited LIDC label-noise budget. We computed pairwise inter-rater agreement with a committed script (`pipelines/pwm_ldct_prep/interrater_metrics.py`) over the full 911-case LIDC `raw_per_reader` release (4 readers/case, 3,643 annotation files). On the primary analysis group

(*anon_1-anon_4*, 677 cases), pairwise detection Cohen’s κ ($\text{IoU} \geq 0.30$ matching) ranged 0.926–0.997 across all six reader pairs, all above the $\kappa \geq 0.60$ calibration gate; linearly-weighted texture κ (1–5 scale) ranged 0.330–0.471, which is lower and is disclosed as a limitation of the reader-sensitive ordinal texture scale (mitigated by ≥ 3 -of-4 majority consolidation). Likert ICC(2,k) is not reported because no Likert series exist in the current release (`annotations/likert/` absent). Top-up annotation on AAPM and Mayo chest cases is planned for v1.0; at that point each top-up annotator’s Cohen’s κ against the LIDC majority-vote calibration set will be reported (the same calibration gate used at onboarding, $\kappa \geq 0.60$). The discordant per-reader labels behind every consensus nodule are preserved in `annotations/raw_per_reader/` so downstream uncertainty-aware methods can access reader-level signal.

Baseline reproductions

To demonstrate that the v0.5 unified loader and Docker preprocessing pipeline support end-to-end training and evaluation of heterogeneous deep reconstruction methods without code modification, we reproduce four published baselines on the v0.5 LIDC training split and report their test-split performance in Table 10. The baselines span method classes: RED-CNN (Chen et al., 2017) (encoder-decoder CNN), a published transformer-based reconstruction method (CTformer (Wang et al., 2023)), a published diffusion-based reconstruction method (CoreDiff (Gao et al., 2023)), and a published unrolled-iterative reconstruction method (LEARN (Chen et al., 2018)). This section is intentionally not a benchmark ranking: relative method comparison is out of scope for a Data Descriptor and is the subject of separate companion work. The role of these reproductions is solely to provide evidence that (i) the v0.5 loader drives heterogeneous published methods to convergence, and (ii) the reported numbers can be reproduced from the released Docker container within the documented per-voxel and floating-point tolerances. Training for v0.5 is planned on the LIDC-only split; AAPM/Mayo training data arrive with the v1.0 roadmap.

Detectability is not a substitute for a reader study; the model observer (CHO with four difference-of-Gaussians channels, no internal noise) is a proxy computed at an agent’s cost rather than a radiologist’s time, and clinical claims remain out of scope. Table 11 pairs the fidelity numbers of Table 10 with a task-based detectability score per the companion charter’s rule that both numbers appear together or neither appears. The task is stated before the observer: a known-location, 2D Gaussian signal of 20 HU peak contrast at $\sigma = 2$ px (SKE), and the noise ROI is constrained to tissue (HU band 10–120, low gradient), so detectability is never measured over anatomy. A fixed Gaussian blur ($\sigma = 1.0$, 5×5) is retained as a permanent control (the blur trap): it reaches a PSNR comparable to or above CTformer at every dose while its CNR falls below the Rose criterion of 3.0 at r010 and r025, i.e. a fidelity gain (or a PSNR parity) with low detectability is a failure of diagnostic content, not a mixed result. The CHO AUC saturates at 1.000 for all methods on this SKE task, so CNR is the discriminating index at the stated task difficulty; task difficulty, observer channels, and internal-noise settings are declared in `baselines/src/pwm_ldct_baselines/observers.py` (TaskSpec) and in the per-model result JSONs under `baselines/results/`. The detectability run used $n = 30$ test slices per dose level as a P0 pilot; the full $n = 764$ detectability sweep is queued behind the same Docker-based evaluation endpoint and will be reported in the companion benchmark work rather than in this Data Descriptor.

Usage Notes

Loading the data

Install the data loader:

```
pip install pwm_ldct_loader
```

Then point it at your local copies of the source distribution (v0.5 loads LIDC; AAPM/Mayo roots are reserved for the v1.0 roadmap):

```
from pwm_ldct_loader import LowDoseCTDataset
train = LowDoseCTDataset(
    root_lidc="/path/to/lidc_idri_dicom/",
    # root_aapm="/path/to/aapm_2016/", # v1.0 roadmap
    # root_mayo="/path/to/mayo_ldct_pd/", # v1.0 roadmap
    split="train", seed=42,
)
sample = train[0]
# sample["full_dose"]    -> Tensor [H, W]
# sample["low_dose"]     -> Tensor [H, W] (simulated at 0.10/0.25/0.50 in v0.5; reference n
# sample["sinogram"]     -> Tensor [views, dets] (where available)
# sample["source"]       -> str: "lidc" (v0.5); "aapm" | "mayo" reserved for v1.0
# sample["annotations"]  -> dict
# sample["metadata"]     -> dict
```

The loader pins random seeds, propagates them to PyTorch / NumPy / CUDA RNGs, and uses deterministic CuDNN settings. The `source` field is exposed so downstream methods can stratify by source if they need to (v1.0+).

Reproducing the baseline results

```
docker run --gpus all \
    -v /path/to/pwm_ldct_1_0:/data \
    -v $(pwd)/output:/output \
    pwm-ldct-red-cnn:v1 \
    eval --dataset /data --split test \
        --out /output/results.json --seed 42
```

The output `results.json` should match the published values within the documented floating-point tolerance on equivalent hardware. A separate community leaderboard that evaluates submissions against this release is in preparation and described elsewhere; it is an optional ecosystem add-on and is not a prerequisite for using the dataset.

Analyses uniquely enabled by v0.5

PWM-LDCT v0.5 is the first release to support the following analyses out of the box:

1. **Cross-vendor generalization.** The LIDC cohort spans four manufacturers (GE, Siemens, Philips, Toshiba), so v0.5 supports leave-one-vendor-out training/evaluation on the harmonized LIDC-only schema, with simulated low dose at 0.10/0.25/0.50 — a cross-vendor analysis no prior single public low-dose CT release enabled on this scale.

2. **Simulation-level low-dose studies.** v0.5 ships simulated low-dose reconstructions at three dose ratios from a committed projection-domain forward model, so downstream methods can train/evaluate low-dose reconstruction without re-acquired pairs.

3. **Annotation-uncertainty studies.** The release preserves the per-reader LIDC annotations alongside the majority-vote ground truth, so methods that want to study label uncertainty have direct access to the raw annotator-level data.

Cross-source consistency studies (LIDC vs AAPM vs Mayo), reference noise-inserted paired-dose comparison, and the two-vendor (GE↔Siemens) real-paired-dose analyses arrive with the v1.0 roadmap. A planned prospective companion dataset extends coverage with prospectively-acquired, multi-site data at ≥ 500 -patient scale, additional vendors (toward Canon and Philips), and new clinical tasks (liver-lesion segmentation, pediatric subset as companion); analyses that depend on those extensions are out of v0.5 scope.

Limitations and bias acknowledgment

v0.5 is a code-and-metadata release on top of the existing public LIDC data. Its limitations are inherited from the underlying source or, where noted, are v0.5-specific scope decisions.

Four-vendor LIDC coverage; real paired-dose data deferred. The LIDC cohort spans four manufacturers (GE, Siemens, Philips, Toshiba), so v0.5 supports genuine cross-vendor (leave-one-vendor-out) evaluation on simulated low dose. It does *not* include real paired-dose acquisitions; the two-vendor (GE vs Siemens) real paired-dose Mayo cohort and the AAPM 2016 reference-noise-insertion comparison are deferred to the v1.0 roadmap, and prospective, multi-site acquisition spanning additional vendors is deferred to the prospective companion dataset.

LIDC-only scope (v0.5) and small real-paired-dose roadmap. v0.5 contains LIDC only (1,010 patients). The combined AAPM (10) + public Mayo (199) real paired-dose cohort planned for v1.0 is 209 source records, or **208 unique patients** after de-duplicating the one patient (L143) shared by both sources; while enough for credentialed evaluation of the aggregate chest task, fine-grained sub-population credentials (e.g., per-vendor $\times$ per-pathology) would be underpowered there too.

No re-acquired low-dose; reduced-dose is simulated. No paired low-dose in v0.5 is a second physical acquisition. For LIDC (full-dose only) we synthesize low-dose via Poisson-thinning at 0.10/0.25/0.50 with a committed projection-domain forward model. Simulation can underestimate true low-dose noise in some regimes (Wang et al., 2020); the v1.0 roadmap will make the simulation-vs-reference (Mayo noise-insertion) discrepancy measurable (Figure 3), and truly re-acquired paired dose is deferred to the prospective companion dataset.

Annotated-task coverage is narrow. The LIDC source is predominantly adult chest scans for lung-cancer screening or nodule evaluation. v0.5 ships harmonized lung-nodule-detection annotations (four-reader raw + majority vote); other tasks on the LIDC subset remain unannotated in v0.5, and abdominal scans with lesion annotations arrive only with the v1.0 roadmap’s Mayo cohort (deferred annotation tasks are also covered by the prospective companion dataset). Pediatric, cardiac, and trauma indications are absent from the underlying sources entirely.

Annotation top-up is deferred to v1.0. The AAPM and Mayo top-up annotation is scoped to the chest lung-nodule task and is planned for v1.0; no top-up annotation is included in v0.5.

No new clinical-quality validation. Because v0.5 reuses existing annotations rather than collecting under a single panel, the inter-rater variability inherited from LIDC is preserved (and reported) rather than minimized. The prospective companion dataset’s campaign will run a single calibrated panel under uniform protocol.

Reliance on the source’s continued availability. If the LIDC hosting institution withdraws the dataset, v0.5’s manifest still records what was included but the underlying scans become unavailable. The mirrored content hash (PhysioNet DOI and Zenodo) provides a permanent record of what v0.5 *was*; it does not guarantee the scans remain downloadable.

Data versioning and persistent identifiers

Each release of PWM-LDCT is uniquely identified by the SHA-256 hash of its `manifest.sha256` file, which itself hashes every released artifact (annotation files, simulated-low-dose images, harmonized metadata, splits, Dockerfiles, and schema). This content hash lets any downstream user verify offline that their local copy of the deposited artifacts is bit-identical to the release, and that shards they regenerate from their own source distributions match the release within the documented per-voxel tolerance, independent of any hosting institution remaining online.

The release is deposited and cited through standard, journal-recognized channels: the assigned **PhysioNet DOI** (10.13026/[**TODO: prefix**]-0.5.0) is the primary citation handle, and a versioned archive of the code and deposited records is mirrored to **Zenodo** (DOI [**TODO: Zenodo DOI**]) and the project’s GitHub release, with the same SHA-256 content hash recorded in each so the descriptor’s reproducibility claims do not depend on any single channel remaining available.

Releases follow MAJOR.MINOR.PATCH semantics: MAJOR breaks schema compatibility, MINOR adds sources or annotation tasks without breaking schema, and PATCH corrects errors (e.g., a scan flagged after PHI re-audit). The v1.0 roadmap (AAPM 2016 + Mayo LDCT-PD) ships as a MINOR release with its own DOI and content hash; the planned prospective companion dataset (below) ships as a separate release with its own identity, not as a patch to this one.

Errata and data-correction policy. Errors (a missed PHI hit, a discordant annotation surviving QA, a reproducibility regression) are reported via GitHub issues (https://github.com/integritynoble/low_dose_CT/issues, label `errata-v0.5`) and batched into PATCH releases that increment the version, regenerate the content hash, and update the deposited record and its mirrors. Each PATCH ships an erratum note (what changed, which scans are affected) and a cumulative `ERRATA.md`. Prior versions remain resolvable by their content hashes, so papers citing an earlier release stay reproducible.

Planned v1.0 roadmap and prospective companion dataset

v1.0 roadmap. The v1.0 roadmap adds the AAPM 2016 (10 patients) and Mayo LDCT-PD (public, 199 patients) sources under the same harmonization schema, with their real paired-dose data (full-dose acquisitions; reduced-dose by Mayo’s validated noise insertion), top-up lung-nodule annotation on chest cases, and cross-source consistency / simulation-vs-reference validations. The v1.0 release ships as a MINOR release with its own DOI and content hash.

A further prospective companion dataset — a separate future release with its own identity and content hash, *not* a revision of PWM-LDCT v0.5 — is planned to address the limitations explicitly tied to the lack of new prospective acquisition. It is planned to add:

- Prospectively-acquired, multi-site paired-dose CT at ≥ 500 -patient scale (UTSW + partner-site clinical data), extending vendor coverage beyond the four vendors already in v0.5 (GE, Siemens, Philips, Toshiba) and the GE + Siemens planned for v1.0, toward Canon and Philips
- Single-panel calibrated radiologist annotations under uniform protocol across all scans
- Liver-lesion segmentation as an additional clinical task on abdominal scans
- Pediatric chest scans (planned as a further companion dataset rather than a patch to this release)
- Extended leave-one-vendor-out evaluation across ≥ 3 vendors (v0.5 already enables 4-vendor LIDC leave-one-vendor-out splits)

The companion dataset’s timing is gated on PI staffing at the coordinating site, IRB approval, and partner-site agreements; the target release window is approximately 12–18 months after this one. PWM-LDCT v0.5 ships independently and is not blocked by the companion’s clinical-acquisition timeline. This roadmap is provided for reader context; it is not part of the v0.5 descriptor’s validated claims.

Code Availability

- Dataset loader: https://pypi.org/project/pwm_ldct_loader/ (Apache 2.0)
- Preprocessing Dockerfile: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/pipelines/ (LIDC pipeline ships with v0.5; AAPM/Mayo pipelines are v1.0 roadmap)
- Annotation artifacts: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/annotations/
- Forward model used for low-dose simulation: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/pipelines/pwm_ldct_prep/lowdose_sim.py (reference implementation; defers to `pwm_core.contrib.modalities.ct_radon` when installed)
- Deposited records + manifest: the PhysioNet release record under DOI 10.13026/[**TODO: prefix**]-0.5.0, archived at Zenodo ([**TODO: Zenodo DOI**]). The manifest’s SHA-256 content hash is recorded in both.

Author contributions

[**TODO: Fill at submission. Use the CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.**]

Competing interests

One or more authors are affiliated with the PWM Protocol Foundation, which contributed the open-source registry tooling used in preprocessing. All data records and code are openly licensed (Table 3) and are fully obtainable, citable, and verifiable through the PhysioNet DOI and the Zenodo archive, so this affiliation confers no privileged access to the release. **[TODO: At submission, each author declares any financial interest in the PWM Protocol Foundation (equity, consulting, or advisory role) or states its absence.]** The authors declare no other competing interests.

Acknowledgments

We thank the LIDC-IDRI release team and the National Cancer Institute for the underlying scans and the existing four-radiologist annotation infrastructure that v0.5 builds upon. We thank the AAPM 2016 Low-Dose CT Grand Challenge and Mayo LDCT-PD release teams for their datasets, which the v1.0 roadmap will add under the same harmonization schema. We thank the PWM Protocol Foundation for the content-addressed registry infrastructure and the open-source maintainers of `pwm_core`, PyTorch, NumPy, PyDICOM, and Tesseract.

References

- AAPM. 2016 NIH-AAPM-Mayo clinic low dose CT grand challenge. <https://www.aapm.org/grandchallenge/lowdosect/>, 2016. Accessed for the PWM-LDCT release.
- S. G. Armato III, G. McLennan, L. Bidaut, et al. The lung image database consortium (LIDC) and image database resource initiative (IDRI): A completed reference database of lung nodules on CT scans. *Medical Physics* 38(2):915-931, 2011.
- Hu Chen, Yi Zhang, Mannudeep K. Kalra, Feng Lin, Yang Chen, Peixi Liao, Jiliu Zhou, and Ge Wang. Low-dose CT with a residual encoder-decoder convolutional neural network. *IEEE Transactions on Medical Imaging*, 36(12):2524–2535, 2017. doi: 10.1109/TMI.2017.2715284.
- Hu Chen, Yi Zhang, Yunjin Chen, Junfeng Zhang, Weihua Zhang, Huaqiang Sun, Yang Lv, Peixi Liao, Jiliu Zhou, and Ge Wang. LEARN: Learned experts’ assessment-based reconstruction network for sparse-data CT. *IEEE Transactions on Medical Imaging*, 2018. arXiv:1707.09636.
- Qi Gao, Zilong Li, Junping Zhang, Yi Zhang, and Hongming Shan. CoreDiff: Contextual error-modulated generalized diffusion model for low-dose CT denoising and generalization. *arXiv preprint arXiv:2304.01814*, 2023.
- Cynthia H. McCollough, Adam C. Bartley, Rickey E. Carter, Baiyu Chen, Tammy A. Drees, Phillip Edwards, David R. Holmes, Alice E. Huang, Farhana Khan, Shuai Leng, Kyle L. McMillan, Gregory J. Michalak, Kristina P. Nunez, Lifeng Yu, and Joel G. Fletcher. Low-dose CT for the detection and classification of metastatic liver lesions: Results of the 2016 low dose CT grand challenge. *Medical Physics*, 44(10):e339–e352, 2017. doi: 10.1002/mp.12345.
- National Cancer Institute. National lung screening trial (NLST) data. <https://cdas.cancer.gov/nlst/>, 2013. Controlled-access distribution via the NCI Cancer Data Access System (CDAS); image data via TCIA.

- 627 National Lung Screening Trial Research Team. Reduced lung-cancer mortality with low-dose
628 computed tomographic screening. *New England Journal of Medicine*, 365(5):395–409, 2011. doi:
629 10.1056/NEJMoa1102873.
- 630 PWM Protocol. PWM Core: A 172-modality engine for verified scientific comput-
631 ing. <https://github.com/integritynoble/pwm>, 2026. Forward model:
632 `packages/pwm_core/contrib/modalities/ct_radon.py`.
- 633 Ray Smith. An overview of the Tesseract OCR engine. In *Ninth International Conference on*
634 *Document Analysis and Recognition (ICDAR)*, pages 629–633, 2007.
- 635 Dayang Wang, Fenglei Fan, Zhan Wu, Rui Liu, Fei Wang, and Hengyong Yu. CTformer: Convolution-
636 free Token2Token dilated vision transformer for low-dose CT denoising. *Physics in Medicine &*
637 *Biology*, 2023.
- 638 Ge Wang, Jong Chul Ye, and Bruno De Man. Deep learning for tomographic image reconstruction.
639 *Nature Machine Intelligence*, 2(12):737–748, 2020. doi: 10.1038/s42256-020-00273-z.
- 640 Dong Zeng, Jing Huang, Zhaoying Bian, Shanzhou Niu, Hua Zhang, Qianjin Feng, Zhengrong Liang,
641 and Jianhua Ma. A simple low-dose X-ray CT simulation from high-dose scan. *IEEE Transactions*
642 *on Nuclear Science*, 62(5):2226–2233, 2015.

Dimension	LIDC-IDRI (Armato III et al., 2011)	AAPM 2016 (AAPM, 2016)	Mayo LDCT-PD (McCollough et al., 2017)	NLST (National Lung Screening Trial Research Team, 2011 ; National Cancer Institute, 2013)	PWM-LDCT v0.5
Scanner-vendor coverage	4 vendors (GE, Siemens, Philips, Toshiba)	Siemens only	GE + Siemens ($\approx 50/50$)	Multi-site, multi-vendor (screening)	4 vendors (LIDC-derived)
Paired full/low dose	No (full-dose only)	Yes, 25% noise-inserted	Yes, 25% noise-inserted	No (single screening acquisition)	No real paired dose; simulated LD at 0.10/0.25/0.50 (roadmap: v1.0 paired)
Raw projections (sinograms)	No	Yes	Yes	No	No (roadmap: v1.0 AAPM + Mayo)
Sample size (patients)	1,010	10	199 (public)	26,722 (LDCT arm)	1,010 (v0.5); 1,218 planned for v1.0
Annotation depth	4-radiologist nodule labels	None as-distributed	Lesion labels on subset	No pixel-level labels (trial outcomes only)	Harmonized multi-task: nodule majority-vote + per-reader (roadmap: top-up + Likert in v1.0)
Dose protocol	Standard clinical dose	Projection-domain noise insertion at 25%	Projection-domain noise insertion at 25%	Fixed low-dose screening protocol (non-paired)	Physically-grounded simulated LD (Poisson-thinning); roadmap: real noise-inserted reference
Unified schema / content addressing	No	No	No	No	Yes (HDF5 shards + SHA-256 manifest)
Access model	Open (NBIA, CC BY 3.0)	Research DUA (Mayo)	Credentialed (TCIA)	Controlled (NCI CDAS)	PhysioNet DOI + Zenodo mirror (records CC BY 4.0)

Table 2: Comparison of PWM-LDCT v0.5 with existing public low-dose CT resources. “Paired full/low dose” refers to same-patient acquisitions or matched low-dose derivatives generated from the patient’s own full-dose projection data; NLST is a screening trial and does not provide paired-dose scans, pixel-level annotations, or raw projections. PWM-LDCT v0.5 harmonizes LIDC-IDRI (its sole source) into a content-addressed, multi-task benchmark; the AAPM 2016 and Mayo LDCT-PD rows describe resources that PWM-LDCT plans to harmonize in v1.0 (roadmap) rather than datasets included in this release. The v1.0 combination of *cross-vendor* (GE + Siemens) *paired-dose* coverage with harmonized annotations, unified schema, and content-addressed versioning is unique to PWM-LDCT among public low-dose CT resources.

Path	Contents	License avail.	/
<i>(i) New data records — deposited in full</i>			
<code>annotations/lidc_majority_vote</code>	Majority-vote consolidation of the four-radiologist LIDC nodule labels	CC BY 3.0	
<code>annotations/raw_per_reader</code>	Per-reader raw labels (pre-majority-vote), for uncertainty studies	CC BY 4.0	
<code>sim_lowdose/lidc/</code>	Simulated low-dose images for the full-dose-only LIDC scans (HDF5), $r \in \{0.10, 0.25, 0.50\}$	CC BY 3.0	
<code>metadata/</code>	Harmonized per-scan <code>metadata.json</code> (acquisition params, HU offsets, kernel, flags)	CC BY 4.0	
<code>splits/{train,val,test}.txt</code>	LIDC patient IDs per split (seed 42)	CC BY 4.0	
<code>manifest.sha256</code>	SHA-256 of every released artifact (the content address)	CC0	
<i>(ii) Access & reproduction code</i>			
<code>pwm_ldct_loader</code> (PyPI)	Unified Python loader (LIDC in v0.5; AAPM/Mayo in v1.0 roadmap)	Apache 2.0	
<code>pipelines/Dockerfile.lidc</code>	Deterministically regenerate harmonized HDF5 (within the released per-voxel tolerance) from the LIDC source's raw DICOM	Apache 2.0	
<code>schema/*.md</code>	Per-record schema, DICOM cleaning spec, DICOM→HDF5 mapping, annotation-QA protocol	Apache 2.0	
<i>(iii) Not redistributed — obtained from source, regenerated locally</i>			
Underlying DICOM scans	Full-dose LIDC images (NBIA)	LIDC CC BY 3.0 / NBIA terms	
AAPM/Mayo scans + derivatives (v1.0 roadmap)	Full/low-dose images and sinograms; pixel derivatives of DUA-restricted scans	Not included in v0.5	

Table 3: PWM-LDCT v0.5 (LIDC-only) release contents, grouped by license and availability. The new data records (annotations, LIDC-derived simulated low-dose, harmonized metadata, splits, manifest) are deposited in full under open licenses; the access-and-reproduction code is Apache 2.0; the underlying DICOM scans are not redistributed — users obtain the LIDC scans from the NBIA gatekeeper and run the Dockerfile pipeline locally to regenerate HDF5 shards matching `manifest.sha256`. AAPM 2016 and Mayo LDCT-PD, their real low-dose pairs, and their pixel derivatives are v1.0 roadmap items and do not appear in v0.5.

Source	Train	Val	Test	Total
LIDC-IDRI (v0.5)	589	216	205	1,010
<i>v1.0 roadmap</i>				
AAPM 2016	4	2	4	10
Mayo LDCT-PD (public)	118	37	44	199

Table 4: v0.5 cohort composition by source and split (LIDC-only). Splits are patient-level and deterministic given the published seed (42); LIDC contributes 1,010 of its canonical 1,018 patients (8 are unavailable in the public distribution used here). AAPM 2016 and Mayo LDCT-PD rows are the v1.0 roadmap plan and are not part of v0.5.

Parameter	LIDC-IDRI (v0.5)
Manufacturer set	GE, Siemens, Philips, Toshiba (670 / 201 / 73 / 66)
Scanner model(s)	heterogeneous, per-scan in <code>metadata.json</code> (4-vendor)
Tube voltage (kVp), typical	120 (813/1,010 of scans; tail 130–140)
Full-dose effective mAs, typical	median 230 (range 0–11,200; IQR 32–943; site-heterogeneous)
Low-dose ratio (paired)	N/A (full-dose only; v0.5 provides <i>simulated</i> low dose at 0.10/0.25/0.50)
Pitch, typical	median 0.94 (recorded on 132 scans; site-heterogeneous)
Slice thickness (mm), thin series	0.60–5.00 (median 2.0) across patients
Reconstruction kernel(s)	heterogeneous (STANDARD, BONE, B30f, LUNG, and 14 further kernels), per scan

Table 5: Acquisition parameters. v0.5 contains LIDC only ($n = 1,010$); the LIDC cells are regenerated from `metadata.json` by the committed extraction script `analysis/extract_acquisition.py` (build provenance: 1010 files, combined SHA-256 20476a8105319e6192eebf233e274a4f6f5f787a574bd605809963f3160df870). The AAPM 2016 and Mayo LDCT-PD columns are the v1.0 roadmap plan and are not part of v0.5. Full per-scan values — including the exact scanner-model string, pitch, kernel, and per-scan effective mAs — are exposed by the loader without imposing a single value. LIDC’s broad multi-vendor coverage (four manufacturers: GE, Siemens, Philips, Toshiba) is a noteworthy strength for cross-vendor generalization studies.

Field	LIDC-IDRI (v0.5)	AAPM 2016 / Mayo LDCT-PD (v1.0 roadmap)
N (patients, unique)	1,010	10 / 199
Age, median (IQR) [yr]	61 (50–70)*	N/A [†] / 62 (51–72)
Age, range [yr]	14–88*	N/A [†] / 17–87
Sex (M / F), n (%)	140 / 140 (28%)*	N/A [†] / 103 / 96 (100%)
BMI, median (IQR) [kg/m ²]	N/A	N/A
Scan-date range	N/A	N/A
Primary indication	Lung-cancer screening + nodule eval	Oncology (abdomen) / Oncology (chest + abdomen)
Geographic origin (sites)	US multi-site (LIDC consortium)	Mayo Clinic, Rochester MN

Table 6: Patient-level demographics. v0.5 contains LIDC only ($n = 1,010$); the AAPM/Mayo columns are the v1.0 roadmap plan and are not part of v0.5. Age, sex, BMI, and scan-date fields are aggregated from the source’s published metadata where available; fields not preserved in a source’s de-identified release are recorded as N/A. The LIDC cells are regenerated from `metadata.json` by the committed extraction script `analysis/extract_demographics.py` (build provenance: 1010 files, combined SHA-256 20476a8105319e6192eebf233e274a4f6f5f787a574bd605809963f3160df870). *LIDC age and sex are recorded only on a subset of patients (age 20%, sex 28%) per LIDC’s variable DICOM-header preservation; the percentages in the Sex row are computed over the patients with known sex. [†]AAPM 2016 patients are fully anonymized (PatientID = “Anonymous”) in the public distribution; age and sex were stripped at de-identification and are not recoverable. BMI and StudyDate are not preserved in any of the source releases. Geographic origin captures the population from which each cohort is sampled and bears on generalizability claims.

Vendor	n_{series}	W_1 vs. aggregate (HU)	Spacing overlap (%)	Thickness overlap (%)
GE Medical Systems	30	114.5	48.3	29.0
Siemens	30	43.4	61.7	74.7
Toshiba	30	85.6	70.0	—
Philips	30	27.5	63.3	57.9

Table 7: Harmonization validation under v0.5 (LIDC-only): HU-histogram 1-Wasserstein distance to the aggregate (clinical window $[-1100, 500]$ HU), pixel-spacing distribution overlap, and slice-thickness distribution overlap, computed on a stratified random sample of 30 series per vendor (120 total, seed 42) from `D:\ZHY\LIDC3DDataset\output_all`. The worst observed vendor pair ($W_1 = 199.3$ HU, GE–Toshiba) sets the 200 HU threshold quoted in the text. Toshiba contributes no thin (≤ 1.5 mm) series, hence the missing thickness overlap; AAPM/Mayo panels are v1.0 roadmap. Regenerated by the temporary validation script `hv_fast.py` (intermediate provenance).

Case	Slice	Attribute	Published	Recomputed
lidc-0128	172	texture (1–5)	3	5
lidc-0128	173	texture (1–5)	3	5
lidc-0128	174	texture (1–5)	3	5
lidc-0128	175	texture (1–5)	3	5
lidc-0615	173	unrecorded cluster	–	6.62 mm (3 readers)
lidc-0007	112	diameter (mm)	21.484	17.188
lidc-0866	73	texture (1–5)	5	3
lidc-0606	197	diameter (mm)	18.246	13.516
lidc-0606	194	diameter (mm)	17.571	14.191
lidc-0622	142	diameter (mm)	21.723	18.480
lidc-0622	137	texture (1–5)	3	5
lidc-0699	234	diameter (mm)	17.619	7.957
lidc-0699	243	diameter (mm)	17.619	1.137
lidc-0699	231	unrecorded cluster	–	10.8 mm (3 readers)
lidc-0804	255	texture (1–5)	3	5
lidc-0804	254	texture (1–5)	3	5
lidc-0804	253	texture (1–5)	3	5

Table 8: Majority-vote reconciliation discrepancies on the 100-patient random sample. Unrecorded-cluster rows mark nodule-sized (≥ 3 mm) annotations agreed by ≥ 3 of 4 readers that are absent from the deposited majority-vote; diameter rows report the largest per-case deviation between the deposited value and the median of contributing readers; texture rows report label-mode differences > 1 step on the 1–5 scale. Reconciled by the temporary script `nodule_mv_reconcile.py` against `annotations/raw_per_reader/`.

Reader group	N cases	Detection κ (range)	Texture κ (range)	Pass ≥ 0.60
anon_1–anon_4 (primary)	677	0.926–0.997	0.330–0.471	detection: all pairs; texture: no
reader1–reader4	21	0.243–0.751	0.164–0.817	mixed
anonymous (numeric IDs)	6	0.748–1.000	0.000–0.444	detection: all pairs*

Table 9: **Inter-rater annotation reliability.** Pairwise reader agreement computed by `pipelines/pwm_ldct_prep/interrater_metrics.py` over the LIDC `raw_per_reader` files (detection Cohen’s κ after IoU ≥ 0.30 matching, per §3.1 of `schema/annotation_qa_protocol.md`; texture κ = linearly-weighted Cohen’s κ on the 1–5 ordinal scale over matched nodules). Reader groups correspond to four non-homogeneous reader-ID naming schemes present in the raw release (*anon*, *reader*, *anonymous*, numeric); groups are therefore reported separately and are not merged. Likert ICC(2,k) is not reported because no Likert series exist in the current release. *The anonymous group’s texture κ includes 0.000 values for three reader pairs, consistent with a reader-ID naming artifact in this small group ($n = 6$); detection agreement in the same group is high.

Model	r010			r025			r050		
	PSNR (dB)	SSIM	LPIPS	PSNR (dB)	SSIM	LPIPS	PSNR (dB)	SSIM	LPIPS
RED-CNN	51.96	0.9893	0.0053	53.65	0.9929	0.0026	54.46	0.9941	0.0020
LEARN	53.99	0.9912	0.0053	57.06	0.9958	0.0024	58.96	0.9974	0.0018
CTformer	43.20	0.9339	0.0476	43.53	0.9374	0.0435	43.64	0.9385	0.0422
CoreDiff	51.59	0.9877	0.0057	53.36	0.9921	0.0025	54.25	0.9934	0.0017

Table 10: **Baseline reproductions.** Test-split PSNR / SSIM / LPIPS for four baselines on PWM-LDCT v0.5 (LIDC-only), per simulated dose level (0.10/0.25/0.50), computed with the released Docker containers. All four baselines were evaluated on the same $n = 764$ test slices. Following the field rule that a fidelity number is reported only beside a task-based detectability number (§4 of the companion charter), Table 11 reports the paired detectability assessment on the same task.

Model	r010		r025		r050		task & observer		
	PSNR	CNR	PSNR	CNR	PSNR	CNR	task	location	obs
RED-CNN	47.66	1.56	50.64	2.68	52.27	3.69	SKE-Gaussian20HU-s2px	known	CHO(DC)
LEARN	48.17	1.34	52.14	2.73	54.94	4.35	SKE-Gaussian20HU-s2px	known	CHO(DC)
CTformer	42.28	0.21	43.06	0.22	43.34	0.22	SKE-Gaussian20HU-s2px	known	CHO(DC)
CoreDiff	—	—	—	—	—	—	weights not loadable in this run		
Gaussian blur (trap)	43.24	1.67	43.32	2.99	43.34	4.04	SKE-Gaussian20HU-s2px	known	CHO(DC)

Table 11: **Paired fidelity + task-based detectability.** The detectability task is signal-known-exactly (SKE): a 2D Gaussian lesion ($\sigma = 2$ px, peak contrast 20 HU) inserted into the low-dose input at a known location, reconstructed by each method, and scored on a tissue-only ROI (HU band 10–120, low gradient) that contains no anatomy. CNR is computed against the Rose criterion of 3.0 (reliable detection); CHO AUC was also computed with four difference-of-Gaussians channels and no internal noise, and saturated at 1.000 for every method on this SKE task, so CNR is the discriminating index. The **Gaussian blur** (fixed $\sigma = 1.0$, 5×5 kernel) is a permanent control: it scores a PSNR comparable to or above CTformer while its CNR falls below the Rose criterion at r010 and r025, demonstrating that a high PSNR does not imply a detectable lesion (the blur trap). Detectability numbers here use $n = 30$ test slices per dose level (subset of the $n = 764$ used above); the full-scale run is part of the P0 observer module delivered in `baselines/src/pwm_ldct_baselines/observers.py`.

